

MANIKANTA PUDI

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PROFESSIONAL SUMMARY

Entry-level Software Engineer with hands-on experience building full stack applications using Django, FastAPI, React and PostgreSQL. Built a point-of-sale system whose billing runs as one atomic transaction with row-level locking, so concurrent sales cannot oversell stock, plus a FastAPI service over PostgreSQL and ChromaDB with 48 tests. Brings 6 months of paid experience writing Python and SQL.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript

Backend: Django, Django REST Framework, FastAPI, Flask, REST API Design, JWT Authentication, Pydantic, Rate Limiting

Frontend: HTML5, CSS3, React 19, Vite, Bootstrap, React Router, Axios, Responsive Layouts

Databases: PostgreSQL, MySQL, Django ORM, SQLAlchemy ORM, Schema Design, Transactions, Row-Level Locking, Query Optimization

DevOps & Testing: Docker, Docker Compose, CI/CD, GitHub Actions, Git, Pytest, Postman, Swagger/OpenAPI, Render, Vercel, Linux

Core CS: Data Structures, Algorithms, OOP, DBMS, Operating Systems, Computer Networks

PROFESSIONAL EXPERIENCE

Data Science Intern | Infosys Springboard

December 2025 - February 2026

Remote, India

- Built Flask REST APIs serving machine learning predictions in real time across 25 materials and 13 product categories.
- Designed a normalized PostgreSQL schema through SQLAlchemy ORM and loaded models into REST endpoints with input validation and error handling.
- Deployed the service on Render with managed PostgreSQL and per-IP rate limiting at 60 requests per minute.

Data Analyst Intern | Techmatrics Solution

January 2025 - June 2025

Hyderabad, India

- Wrote Python and complex SQL using multi-table JOINS, CTEs and window functions to process 6.7M+ records, automating recurring reporting workflows.
- Optimized queries against PostgreSQL, reducing execution time and eliminating manual data preparation steps.
- Built modular, reusable Python functions for validation, cleaning and aggregation used across reporting pipelines.

PROJECTS

Anvil Hardware - Point-of-Sale and Inventory System | [GitHub](#) | [Live App](#)

July 2026

Python, Django 5.1, Django REST Framework, PostgreSQL 17, React 19, Vite, Bootstrap, JWT, Pytest

Django REST API over PostgreSQL with a React counter interface built for keyboard-only billing, deployed on Render and Vercel.

- Built billing as one atomic transaction locking product rows with SELECT FOR UPDATE so concurrent sales cannot oversell stock, sorting product IDs to enforce a single lock order.
- Designed 7 tables across 3 Django apps with PROTECT on transactional foreign keys, JWT authentication, two server-enforced roles, and server-side search and pagination.
- Built the counter screen for keyboard-only billing with 400ms debounced search and printable invoices via print CSS.

Scholarship Eligibility Assistant - FastAPI Retrieval Service | [GitHub](#) | [Live App](#)

August 2026

Python, FastAPI, PostgreSQL, ChromaDB, React 19, Vite, Pytest, Render, Vercel

REST service answering eligibility questions over 44 scholarship schemes, exposing its own evaluation metrics and query-log trends as endpoints.

- Modelled eligibility rules as typed SQL columns filtered before vector search, cutting ineligible candidates reaching the model from 81% to zero.
- Wrote 48 tests and logged every query with its retrieval score, abstention reason and latency, making quality drift visible after deployment.
- Kept embeddings in PostgreSQL and loaded ChromaDB from them at startup, so the index survives free-tier restarts.

CERTIFICATIONS & ACHIEVEMENTS

- Codegnan - Full Stack Python Development Certification (Ongoing, since June 2026).**
- Research Publication - Off-Road Terrain Identification using YOLOv8, Journal of Electrical Systems (December 2023).**
- SQL (Intermediate) Certification - HackerRank (March 2025).**

EDUCATION

Bachelor of Engineering in Computer Science (Artificial Intelligence & Machine Learning)

2021 - 2025

Chandigarh University, Mohali | CGPA: 7.98/10

Relevant Coursework: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks.